

 सत्यमेव जयते	भारत सरकार दूरसंचार विभाग दूरसंचार अभियांत्रिकी केंद्र Government of India Department of Telecommunications Telecommunication Engineering Centre K.L. Bhawan, Janpath, New Delhi- 110001	 ISO 9001:2015
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No. 26-1/2022-FA/TEC

Date: 31.08.2023

Subject: Minutes of Seventh meeting of National Working Group (NWG)-15 corresponding to ITU-T Study Group (SG)-15 held on 18.08.2023.

The seventh meeting of National Working Group (NWG)-15 corresponding to ITU-T Study Group 15 regarding "*Networks, Technologies and Infrastructures for Transport, access & home*", reconstituted for the Study Period 2022-24, was held on 18.08.2023 under the chairpersonship of Sh. Rakesh Desai, DDG(FA), TEC through video-conferencing. List of participants is placed at Annexure-A.

2. At the outset, Sh. Deepak Pathak, Director (FA) TEC & Convener of NWG-15, welcomed all the participants and briefly explained about the discussions held during the previous meetings.

3. Sh. Rakesh Desai, Chairman NWG-15 & DDG (FA) TEC, greeted all the members and welcomed all participants especially new members from IIT Madras viz. Prof Deepa Venkitesh, and other academia. He highlighted past contributions (5) of NWG-15 in the current study period 2022-24 of ITU-T SG15 and further informed that the next meeting of ITU-T SG15 is scheduled from 20th Nov to 1st Dec 2023 at Geneva. He also requested the members to attend online correspondence discussion meetings called by the concerned Rapporteurs between August to October 2023.

After the introduction, Sh. Rakesh Desai requested the members who have submitted draft contributions to present their contributions for further deliberations and also requested other members to actively participate in the discussions on various contributions.

4. Contribution on Distributed Acoustic Sensing (DAS), one of the technologies for Distributed fibre optic sensing (DFOS) systems (Q6/15)

Sh. Sudipta Bhaumik, M/s STL informed that Rapporteur Q6/15, in the meeting dated 25-27 July 2023, requested the members to start a new work item on DFOS limiting its scope to topics under the responsibility of Q6/15. Accordingly, M/s STL has prepared a contribution on Distributed Acoustic Sensing (DAS). Sh. Ashit Padhi from M/s STL, during his presentation, explained the problems faced with multiple cuts in OFC network of various telecom operators, challenges involved in manual monitoring of the system and tracing of fibre after 10-15 years later after laying. To solve this problem, M/s STL proposed a contribution on - Distributed Acoustic Sensing technology in order to avoid any interruption of telecom services and to reduce the attenuation loss over the period of time including digitization of network with tracing capabilities.

This contribution is very well appreciated by other members and members also gave various suggestions / inputs to consider incorporating in the draft contribution. The NWG-15 requested M/s STL to prepare a draft contribution as per the ITU-T prescribed format by 15th September 2023 and circulate the same among members for their inputs / suggestions so that

this could be discussed in the next meeting of NWG-15 before submitting to ITU-T SG15.
(Action by: M/s STL Team viz. Sh. Sudipta Bhaumik, Sh.Ashit Padhi, Dr. Nagaraju, Sh. Abhishek Sati)

5. Questionnaire on Outdoor Optical Infrastructures sharing between multiple operators (Q7/15)

Sh. Rakesh Desai, presented the contribution prepared on practical and technical aspects of the Optical Infra Sharing between multiple operators. He stated that the responses to the Questionnaire have been prepared in consultation with CS Division of DOT HQ and with some of the TSPs/ ISPs. As draft is almost ready and last date for uploading the responses to the questionnaire is 30th Sept 2023, members are requested to give their inputs by 5th September 2023. Sh.Desai stated that finalised responses to the questionnaire will be uploaded on ITU-T portal within the prescribed time on approval of the competent authority.

(Action By: Sh.Rakesh Desai / Ms.Mudita)

6. Draft TR.fcs- Technical Report on Optical fibres, cables and systems (Q5/15)

Sh. Sudipta Bhaumik, M/s STL & Vice-Chair of WP2/SG15, highlighted that a technical paper on Optical fibres, cables and systems has been published by WP2 of SG15. This technical paper is a compilation of all the ITU recommendations and supplements. It has total 11 chapters and needs to be examined and edited, accordingly. In next meeting, Chapter 1, Chapter 2 and Chapter 3 will be discussed. Sh. Rakesh Desai stated that this technical document will be an excellent compendium of OFC and systems and requested the members to provide inputs for the different chapters. He further pointed out that members may also take guidance by this technical report and if there is any gap, they may propose contributions suggesting technical solutions to such gaps. Link of Technical paper has already been shared to the members.

(Action By: All members)

7. Revision of L.100: Optical fibre cables for duct and tunnel application (Q5/15)

Ms. Mudita Chandra, ADET (Tx), during her presentation, mentioned that this contribution is for the revision of ITU-T recommendation of L.100 (Q5/15) and other contributors are Sh.Vijay Dixit, Dir (Tx) TEC and Sh. Sudipta Bhaumik, M/s STL. Ms. Mudita explained the contribution to the members, stating that a new clause for Electrical continuity is included and some texts are added in relevant clauses for 24 F / 48 F OFC based on TEC existing Standards. She also proposed to add a new informative Appendix I for examples of cables for duct and tunnel applications.

Sh.Desai suggested to incorporate other size OFC like 96F/248F, etc so that we can showcase good amount of work as well as we can know similar best practices / technologies in other countries. The contributors are requested to prepare a draft contribution by 15th September, 2023 considering the inputs given by the members and related technical resources and circulate the same among members for their inputs / suggestions so that this could be discussed in the next meeting of NWG-15 before submitting to ITU-T SG15.

(Action by: Ms. Mudita Chandra, Mr Vijay Dixit, Sh. Sudipta Bhaumik)

8. Draft new Recommendation G.fso (Q6/15)

Sh. Rakesh Desai referred to the discussions in the 6th meeting of NWG-15 in this matter wherein the contributors of the work-item-FSOC (Q6/15) were requested to prepare afresh contribution as well as to explore the possibility of merging of Indian Contribution to the agreed work-item submitted by MIIT, China. He apprised the group that Rapporteur of Q6/15, during the discussions on the contribution G.fso (WD06-24 and WD06-25) in a meeting dt. 25-27 July 2023, requested the contributors to further improve upon the contribution considering the observations conveyed in the Q6/15 meeting and further invited the editor, Mr

Wei (China Unicom), to submit a new Recommendation draft accordingly. Sh.Desai therefore requested the NWG-15 contributors of the work-item-FSOC (Q6/15) to furnish their inputs to the editor of G.fso if new work-item is not proposed in this subject.
(Action By : S/Sh. Abdul Kayum/ Avadhesh Singh/ Rakesh Goyal)

9. Sh. Rakesh Desai informed the group about ongoing discussions on, i) proposed methodology and schedule for collecting data on fibre statistics (Q2/15) and ii) proposal on statistical chromatic dispersion property of G.652 and G.657 fibres (Q2,5,6/15). He requested the subject experts to look into the documents related to these discussions and suggest possible contributions.

10. Draft revised Recommendation L.360: "Operations support system requirements for infrastructure and network elements management using Identification data (ID) technology" (Q7/15)

Sh.Rakesh Desai explained the participants regarding Q7/15 correspondence discussions on the Draft revised Recommendation L.360: "Operations support system requirements for infrastructure and network elements management using ID technology". He highlighted the possibility of contributions in this work-item as an Addendum to incorporate Indian Experience on the lines of Italian experience at Appendix I regarding RFID tag solution for telephony poles and additional texts to Appendix II - Advanced technologies used for network infrastructure management in view of the fact that many Indian ISPs, through software tool like Virtual Network Expert (VNE), are already monitoring their FTTX networks as well as enhancing network performance & transformation and customer's experience. He further mentioned that draft contribution in this regard (VNE tool) is under preparation and will be circulated among members for their inputs / suggestions by **15th September 2023**. He requested the members from TSPs/ISPs to share their inputs on above-stated Indian experience for Appendix I of L.360.

(Action By: Sh.Rakesh Desai, Members from TSPs/ISPs)

11. Use of Standards Coordination Portal

Sh. Rakesh Desai informed that more than 35 members have already registered on 'Standards Coordination Portal' of TEC and further requested other members to register on 'Standards Coordination Portal' of TEC (<https://tec.gov.in/scp/>) and start familiarizing with the features of the portal. He further mentioned that important circulars, meeting notices and minutes of meeting will be posted on this portal only in due course.

12. Formation of NWP for NWG-15.

Chairman NWG-15 proposed forming of National Working Parties (NWP) under NWG-15 on the lines of ITU-T WPs to enhance stakeholder involvement from industry, academia, and R&D institutions in ITU standardization activities. These NWPs consist of members from diverse domains, will coordinate specific technical work under NWG-15. Sh.Desai further conveyed that the NWPs may be led by members from industries, start-ups, academia, and researchers. Participants were requested to volunteer for the Chair, Vice-chair and member positions of the proposed NWPs under the NWG-15. The group agreed to constitute two NWPs as of now, one for Q3/15 & 6/15 and another for Q5/15 & 7/15. Sh.Desai informed that he had already initiated discussion with some of the active members and also perused the profiles of some of the members. Based on it, he suggested following constitution for the proposed two NWPs: -

i) National Working Party- NWP 1/15: Q3/15, 6/15

Free Space Optical Communication, Light Communication and in-premises Technologies and related access applications

Position	Name	Type	Organisation
Chair	Mr Deepak Solanki	Start-Up (Industry)	M/s Velmenni Research & Development Pvt. Ltd
Vice-Chair	Dr Gyandeep Verma	Academia	IIT Delhi
Vice-Chair	Mr Hardik Soni	Industry	M/s Nav Wireless Technologies Pvt Ltd.
Member	Mr Hardik Raval	Industry	M/s Nav Wireless Technologies Pvt Ltd.
Member	Mr Vineet Sharma	Start-Up (Industry)	M/s Velmenni Research & Development Pvt. Ltd
Member	Dr Sonali	Industry	M/s HFCL Limited
Member	(To be nominated by Prof Deepa Venkitesh)	Academia	Indian Institute of Technology Madras
Member	Mr Rakesh Goyal	Govt	TEC

ii. National Working Party (2/15): Q5/15, 7/15

Optical fibres and cables and optical physical infrastructures

Position	Name	Type	Organisation
Chair	Sh Sudipta Bhaumik	Industry	M/s Sterlite Technologies Limited
Vice-Chair	Dr Deepak Jain	Academia	IIT Delhi
Vice-Chair / Member	(To be nominated by Prof Deepa Venkitesh)	Academia	Indian Institute of Technology Madras
Member	Mr Roshan Kumar	Industry	M/s Sterlite Technologies Limited
Member	Sh.Guharajan Shivakumar / Sh. Rahul Dhar	Industry	M/s Aprecomm Private Limited, Bangalore
Member	Ms. Mudita Chandra	Govt	TEC

NWG-15 members, who are willing to be part of the NWP 1/15 & 2/15, are requested to send their nomination to the Chair of NWG-15 **by 10th September 2023**. Suggestions / Inputs are also invited for the above-proposed NWP and formation of new NWPs for left out questions of SG-15 **by 10th September 2023**.

13. Participation of academic and research institutes in NWG-15

Sh. Rakesh Desai, Chairman NWG-15 expressed his concern stating that the level of participation from academia in NWG-15 meetings fell short of expectations. He urged all members to identify and nominate relevant experts, including subject experts from academia and practitioners who are willing to contribute in standardization area, to join NWG-15.

14. Involvement of start-ups and other industries (related to telecom field) in the ITU-T processes.

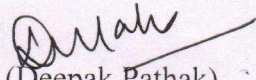
In order to foster greater involvement from start-ups, the Chairman, NWG-15 encouraged members to provide their views and inputs via email.

15. Sh. Desai raised concerns about the irregular attendance of several members in NWG-15 meetings. He urged all the members to participate actively in group activities. He further requested all members to provide ideas or suggestions to improve active participation.

16. The key action points emerged from the meeting are as under: -

- i. Contribution on Distributed Acoustic Sensing (DAS) (Q6/15): As discussed at para 4 above, Action by: M/s STL Team viz. Sh. Sudipta Bhaumik, Sh. Asit, Dr. Nagaraju, Sh. Abhishek Sati, **Timeline: 15th September 2023.**
- ii. Questionnaire on Outdoor Optical Infrastructures sharing between multiple operators (Q7/15): As discussed at para 5 above, inputs from all members **by 5th September 2023.** Finalization of responses by Sh. Rakesh Desai / Ms. Mudita **by 15th September 2023.**
- iii. Draft TR.fcs- Technical Report on Optical fibres, cables and systems (Q5/15): - As discussed at para 6 above, Action by: All Members.
- iv. Revision of L.100: Optical fibre cables for duct and tunnel application (Q5/15): As discussed at para 7 above, Action by: Ms. Mudita Chandra, Mr Vijay Dixit, Sh. Sudipta Bhaumik, **Timeline: 15th September 2023.**
- v. Draft new Recommendation G.fso (Q6/15): As discussed at para 8 above, Action by: S/Sh. Abdul Kayum/ Avadhesh Singh/ Rakesh Goyal.
- vi. Draft revised Recommendation L.360: "Operations support system requirements for infrastructure and network elements management using ID technology" (Q7/15): As discussed at para 10 above, Action by: Sh. Rakesh Desai, Members from TSPs/ISPs; **Timeline: 15th September 2023.**
- vii. Formation of NWPs for NWG-15:- As discussed at para 12 above, NWG-15 members are requested to send their nomination and Suggestions / Inputs to the Chair of NWG-15 **by 10th September 2023.**
- viii. NWG-15 Members may please register on 'Standards Coordination Portal' of TEC (<https://tec.gov.in/scp/>) and start familiarizing with the features of the portal.
- ix. Members may study all the questions of ITU-T SG15 and come forward to increase number of Indian contributions in ITU-T SG15.

17. The meeting ended with the vote of thanks to the chair.


(Deepak Pathak)

Convener, NWG-15 & Director (FA), TEC

To,
All the participants / Members of the NWG-15.
Copy to: Sr.DDG, TEC for kind information.

List of Participants

1. Sh. Rakesh Desai, DDG (FA) & Chairman/NWG-15, TEC
2. Sh. Sudipta Bhuamik, M/s STL, Mumbai
3. Prof Deepa Venkitesh, IITM and other academia from IIT Madras
4. Sh. Rajeev Kumar Kaushik, PGM (NWP-CFA & CDN-I), BSNL CO, New Delhi
5. Sh. Ashit Padhi, M/s STL
6. Dr. Nagaraju Bezawada, M/s STL
7. Sh. Abhishek Sati, M/s STL
8. Sh. Arvind Kansal
9. Sh. Gyan Deep Verma, IIT, Delhi
10. Sh. Rajeev Chaturvedi, Dura-Line India Ltd.
11. Sh. Vijay Madan, TSDSI
12. Sh. Alpesh Rawal, Huber Suhner Electronics
13. Sh. Deepak Pathak, Director (FA-I) TEC and convener NWG-15
14. Sh. Vijay Dixit, Director (Tx), TEC
15. Sh. Brajesh Kumar, Director (FA-II), TEC
16. Ms. Mudita Chandra, ADET (Transmission), TEC
17. Sh. Mohit Aggarwal
18. Sh. Rakesh Goyal, ADG (QT), TEC
19. Sh. Sanjay Bhardwaj, TEC
20. Sh. Ragunathan J, AD (FA), TEC.
